

Practical 1

Example 1: Create a Simple ASP.NET Core Web Application

Step 1: Open Visual Studio 2019

- Launch Visual Studio 2019.
- Click **Create a new project**.

Step 2: Select Project Template

- Search for **ASP.NET Core Web App**.
- Select it.
- Click **Next**.

Step 3: Configure Project

- Project Name: HelloWebApp
- Location: Choose a folder
- Click **Create**

Step 4: Choose Framework

- Select **.NET Core**
- Select desired framework version
- Choose **Web Application (Model-View-Controller)** or **Empty**
- Click **Create**

Visual Studio creates the project automatically.

Example 2: Display "Hello World"

Open:

Controllers/HomeController.cs

Modify the action:

using Microsoft.AspNetCore.Mvc;

```
namespace HelloWebApp.Controllers
{
    public class HomeController : Controller
    {
        public IActionResult Index()
        {
            return Content("Hello World from ASP.NET Core!");
        }
    }
}
```

Save the file.

Step 5: Run the Application

Press:

F5

or click:

IIS Express

Browser opens automatically.

Output:

Hello World from ASP.NET Core!

Example 3: Create a Simple Form

HomeController

using Microsoft.AspNetCore.Mvc;

namespace HelloWebApp.Controllers

```
{
    public class HomeController : Controller
    {
        public IActionResult Index()
        {
            return View();
        }

        [HttpPost]
        public IActionResult Index(string name)
        {
            ViewBag.Message = "Welcome " + name;
            return View();
        }
    }
}
```

Create View

Open:

Views/Home/Index.cshtml

Replace contents:

```
<form method="post">
    <label>Name:</label>
    <input type="text" name="name" />
    <button type="submit">Submit</button>
</form>
```

```
@if(ViewBag.Message != null)
{
    <h2>@ViewBag.Message</h2>
}
```

Run application.

Enter a name and click **Submit**.

Output:

Welcome John

Example 4: Test the Application

Manual Testing

Verify:

- ✓ Application starts
- ✓ Home page loads
- ✓ Form accepts input
- ✓ Correct message displayed

Example 5: Create Unit Tests

Add Test Project

1. Right-click Solution
2. Add → New Project
3. Select **xUnit Test Project**
4. Name it:

HelloWebApp.Tests

Install Package

dotnet add package Microsoft.AspNetCore.Mvc.Testing

Sample Test

```
using Xunit;
namespace HelloWebApp.Tests
{
    public class SimpleTests
    {
        [Fact]
        public void AdditionTest()
        {
            int result = 5 + 5;
            Assert.Equal(10, result);
        }
    }
}
```

Run Tests

- Open **Test Explorer**
- Click **Run All**

Result:

Passed

Example 6: Debugging

Set a breakpoint:

ViewBag.Message = "Welcome " + name;

Click left margin beside the line.

Run:

F5

Submit form.

Execution pauses at the breakpoint, allowing you to inspect variables.

Common Problems and Fixes

NuGet Restore Error

Unable to load the service index for source

<https://api.nuget.org/v3/index.json>

Try:

dotnet restore

or

dotnet nuget locals all --clear

Also ensure internet access and that NuGet.org is enabled in Visual Studio:

Tools → NuGet Package Manager → Package Sources

Build Error NETSDK1004

project.assets.json not found

Run:

dotnet restore

Then rebuild:

Build → Rebuild Solution